Assignment: What-if Analysis and Excel Solver Optimization

Sheet	Topic	Student Task	Method/Tool	Deliverable
1_RangeNames_WhatIf	Range names + model audit	Create and use range names for a subscription profit model	Name Manager / Create from Selection	Named range list + formulas
2_DataTables	One- and two-variable data tables	Evaluate how price and marketing growth change monthly profit	Data Table	Completed sensitivity tables + observations
3_GoalSeek_Scenarios	Goal Seek + Scenario Manager	Find required add-on conversion and compare operating scenarios	Goal Seek / Scenario Manager	Target solution + scenario summary
4_Solver_Linear	Linear optimization	Allocate a marketing budget across channels	Simplex LP	Optimal spend, signups, slack, recommendation
6_Solver_Nonlinear	Nonlinear optimization	Optimize rental prices when demand changes with price	GRG Nonlinear	Optimized prices, nonlinear interpretation

Important reminders

Use yellow cells for decisions or editable assumptions. Green cells are outputs. Do not overwrite formulas unless instructed.
For Solver sheets, write down the objective cell, changing cells, constraints, solving method, and final recommendation.
This assignment is intentionally different from the in-class exercise: it uses an e-bike rental and service expansion case.
Submit the completed Excel workbook and short written answers requested in the instruction file.

NorthStar E-Bike Rentals: Monthly Subscription Profit Model

Input	Value
Base monthly subscribers	2500
Monthly subscription price	$42
Variable service cost / subscriber	$16
Add-on conversion rate	25%
Add-on monthly fee	$12
Add-on variable cost / user	$3
Monthly fixed cost	$65,000
Marketing growth rate	8%
Churn reduction effect	2%

Required Range Name	Current Reference	Purpose
Base_Subscribers	B4	Initial subscribers before growth/churn effect
Monthly_Price	B5	Main subscription price
Service_Cost	B6	Variable cost per subscriber
Addon_Rate	B7	Percent of subscribers buying add-on
Addon_Fee	B8	Add-on fee per customer
Addon_Cost	B9	Add-on variable cost
Fixed_Cost	B10	Monthly fixed cost
Growth_Rate	B11	Demand growth assumption
Churn_Reduction	B12	Demand lift from lower churn

Output	Value
Forecast subscribers	2754
Add-on customers	689
Subscription revenue	$115,668
Add-on revenue	$8,262
Total revenue	$123,930
Service variable cost	$44,064
Add-on variable cost	$2,066
Total variable cost	$46,130
Monthly operating profit	$12,801
Contribution margin before fixed cost	$77,801

Student Range Name Checklist	
1	Create names from the input labels or use the exact names listed above.
2	Replace at least three formulas in B16:B25 with range names.
3	Add a note explaining why range names make the model easier to audit.
4	Delete or correct any invalid/duplicate names in Name Manager.

Range names make it easier to read formulas because you can tell what the cell is for instead of just B4 or B11.

Sensitivity Analysis with One- and Two-Variable Data Tables

Base Assumptions	Value
Base monthly subscribers	2500
Monthly price	$42
Variable service cost	$16
Add-on conversion rate	25%
Add-on fee	$12
Add-on variable cost	$3
Fixed cost	$65,000
Marketing growth rate	8.0%
Churn reduction effect	2.0%

Output	Value
Forecast subscribers	2754
Add-on customers	688.5
Total revenue	$123,930
Total variable cost	$46,130
Monthly operating profit	$12,801

variable Data Table: Monthly Price vs F

Monthly Price	
$34	-9231.5
$36	-3723.5
$38	1784.5
$40	7292.5
$42	12800.5
$44	18308.5
$46	23816.5
$48	29324.5
$50	34832.5
$52	
$54	

wo-variable Data Table: Price and Marketing Growth vs Pro

	0.0%	5.0%	10.0%	15.0%	20.0%
$36	-8262.5	####	-2589	248.1	3085
$40	1937.5	5284	8631	11978	15325
$44	12137.5	####	19851	23708	27565
$48	22337.5	####	31071	35438	39805
$52	32537.5	####	42291	47168	52045

Student response prompts		
1	Which monthly price produces the highest profit in the one-variable table?	$50, it makes the most profit
2	Under which price and growth combination is profit highest in the two-variable table?	highest price ($50) and highest growth rate (0.2) together give the most profit
3	What tradeoff does the model show between price and subscriber growth?	higher price = more profit per person but if growth doesn't happen it doesn't help as much. lower price could maybe get more customers but each one makes less money

Goal Seek and Scenario Manager: Subscription Operating Profit

Input	Value
Base monthly subscribers	2500
Monthly price	$42
Variable service cost	$16
Add-on conversion rate	54.1%
Add-on fee	$12.00
Add-on variable cost	$3
Fixed cost	$65,000
Marketing growth rate	8%
Churn reduction effect	2.0%
Target monthly operating prc	$20,000

Output	Value
Forecast subscribers	2754
Add-on customers	1489.0878
Total revenue	$133,537
Total variable cost	$48,531
Monthly operating profit	$20,006
Profit gap to target	$6

Goal Seek Task		
Set cell	Target Monthly operating profit	20000
By changing cell	Add-on conversion rate	0.5407
Interpretation	the conversion rate would need to be about 54%, that seems too high to actually happen since its more than double what it is now	

Scenario Manager / Scenario Summary							
Scenario	Monthly price	Variable cost	Add-on conversion	Fixed cost	Growth rate	Operating Profit	Comment
Conservative	$39	$18	18.0%	$70,000	2.0%	-$11,165	loses money
Base	$42	$16	25.0%	$65,000	8.0%	$12,801	this is what we have now
Growth Push	$45	$17	35.0%	$78,000	18.0%	$15,730	good profit but depends on a lot of stuff going right
Premium Positioning	$50	$18	28.0%	$72,000	6.0%	$21,308	highest profit out of all of them

I'd go with Premium Positioning since it has the highest profit. the risk is it depends on people actually paying the higher price, if they don't sign up because its too expensive it might not work out.

Linear Optimization: Allocate Launch Marketing Budget

Channel	Paid Search	Social Video	Campus Ambassadors	Referral Credits	Local Partnerships
Spend decision ($000s)	35.0	30.0	0.0	20.0	5.0
Expected signups per $1K	18.0	14.0	10.0	22.0	12.0
Staff hours per $1K	1.0	0.8	1.5	0.6	1.2
Minimum spend ($000s)	10.0	8.0	0.0	5.0	0.0
Maximum spend ($000s)	35.0	30.0	25.0	20.0	25.0

Metric	Value
Total expected signups	1550
Total spend ($000s)	90
Total staff hours	77
Digital spend ($000s)	65

Constraint	Limit
Budget ($000s)	90
Staff hours	80
Digital spend at least	35

Solver Setup		
Set Objective	B12	Maximize expected signups
Changing Variable Cells	B4:F4	Marketing spend by channel
Constraints	B13 <= E12; B14 <= E13; B15 >= E14; B4:F4 >= B7:F7; B4:F4 <= B8:F8	
Solving Method	Simplex LP	
Question 1	Which channel receive	Paid Search gets the most money ($35k) because its a good mix of high signups and its allowed to spend that much
Question 2	Which constraint is bin	the budget ($90k) is all used up so thats binding. staff hours has a little room left (77 out of 80)
Question 3	What happens if staff h	nothing changes really, because budget runs out before staff hours do
Final Interpretation	Provide a business recc	spend most of the budget on Paid Search, Social Video, and Referral Credits since they give the most signups per dollar

Nonlinear Optimization: E-Bike Rental Pricing with Price-Sensitive Demand

Plan	Quick Ride Pass	Commuter Plus	Weekend Family	Corporate Fleet
Base price	$12	$35	$79	$220
Optimized price decision	$14	$52	$115	$320
Variable cost	$4	$12	$30	$90
Base monthly demand	4200	1700	620	180
Price sensitivity	1.4	1.15	0.9	0.65
Demand at optimized price	3385	1078	442	141
Revenue	$47,386	$56,069	$50,854	$45,149
Contribution margin	$33,847	$43,130	$37,588	$32,451
Minimum allowed price	$9	$25	$60	$175
Maximum allowed price	$18	$52	$115	$320

Metric	Value
Total revenue	$199,459
Total contribution margin	$147,016
Total demand	5046
Average optimized price	$125

Constraint	Limit
Total demand must be at least	5000
No plan demand below	100
Commuter price >= Quick +	18
Weekend price >= Commuter +	30
Corporate price >= Weekend +	100

Solver Setup		
Set Objective	B18	
To	Max	
Changing Cells	B5:E5	
Constraints	B5:E5 between B12:E13; B19 >= E17; B9:E9 >= E18; C5 >= B5+E19; D5 >= C5+E20; E5 >= D5+E21	
Method	GRG Nonlinear	
Question 1	Which price moved most from base price?	Corporate Fleet moved the most, from $220 to $320
Question 2	Why is this model nonlinear?	its nonlinear because demand changes based on price using an exponent, not just a straight line
Question 3	Would maximizing revenue give the same answer as maximizing contribution margin?	no, maximizing revenue would lower the Quick Ride price more to get extra customers, but that doesn't always mean more profit since it costs money to serve those customers too
Question 4	Test one sensitivity scenario by changing a price sensitivity value.	I changed Quick Ride's price sensitivity from 1.4 to 1.6 and the optimal price went down to about $10.67, and total profit went down a little too